

Behavior

Toward a Scientific Theory of Computational Behavior

Cross-disciplinary foundational research paper

Abstract

Behavior is indispensable across psychology, ethology, cognitive science, economics, reinforcement learning, cybernetics, robotics, systems theory, human-computer interaction, and philosophy of action, yet the term refers variously to movement, response, output, action, policy, performance, conduct, and organism-environment coordination. This paper asks what constitutes behavior and how it should be represented scientifically without dependence on a particular interface or technology.

The review finds no single accepted definition, but it identifies a stable analytical core. Behavior is best treated as a bounded, temporally extended trajectory of observable or instrumentally recoverable changes attributable to a system in relation to its environment. Behavior is not identical to state, because it concerns change or persistence across time; not identical to action, because action normally adds agency, organization, or purposive interpretation; not identical to interaction, because interaction requires reciprocal constraint between distinguishable systems; and not identical to a policy, strategy, intention, or goal, which are explanatory or latent structures rather than observations.

Existing theories are sufficient for most component problems but lack a shared evidence-preserving representation spanning human, animal, and artificial systems. The paper therefore proposes a typed trace framework rather than a new behavioral primitive. It separates raw records, detected events, attributed behavioral events, episodes, patterns, actions, strategies, and latent explanations. It supplies a behavior ontology, behavioral grammar, trace model, evidence hierarchy, taxonomy, annotation contract, and falsification criteria. Its central claim is that scientifically useful behavioral representation requires explicit system boundaries, observation functions, temporal segmentation rules, attribution criteria, context, uncertainty, and a prohibition against treating latent explanations as direct observations.

1. Problem and Scope

Computational interaction research frequently treats clicks, commands, gaze shifts, messages, state changes, task completion, and model outputs as behavior. These records are not equivalent. Some are environmental measurements, some are system transitions, some are experimenter-defined events, and some are interpretations of agency, purpose, belief, or strategy. Without a theory of behavior, interaction traces risk becoming collections of implementation-specific events whose explanatory status is unclear.

The problem is not solved by defining behavior as whatever a system does. That formulation is too broad to distinguish behavior from internal dynamics, passive displacement, noise, or arbitrary state change. Conversely, limiting behavior to intentional human action excludes organisms, robots, adaptive software, reflexes, omissions, and learned but non-deliberative conduct. A useful account must remain implementation-independent while retaining empirical discipline.

The paper evaluates whether established theories already supply a sufficient account. The objective is not a universal metaphysical definition. It is a minimal scientific architecture adequate for representing behavioral evidence across computational interactions.

2. Method and Standards

The review compares explanatory structures rather than recounting disciplinary histories. Primary and foundational works were prioritized: Dewey on the reflex arc; Skinner on operant behavior; Tinbergen on ethological explanation; Rosenblueth, Wiener, and Bigelow on behavioral classification and purpose; Ryle and philosophy of action on dispositions and agency; Newell and Simon on problem-solving protocols; ecological and situated approaches on organism-environment coordination; reinforcement learning on policies and trajectories; and HCI on task performance, action sequences, and interaction logs.

For each tradition, the analysis identifies: the entity to which behavior is attributed; the temporal unit; the role of environment; whether purpose or intention is required; the observational basis; the explanatory mechanism; and the treatment of latent variables. Candidate distinctions are retained only when counterexamples show that collapsing them would generate false inferences. This follows the project's canon: define vocabulary, separate observation from interpretation, prefer mechanisms, require operationalization and falsifiability, and leave reusable artifacts.

3. Major Traditions

Tradition	Behavioral object	Typical evidence	Strength and limit
Behaviorism	Environmentally related responding; operants and response classes	Observable response and contingencies	Strong operational discipline; weak on representation, distributed activity, and observer construction.
Ethology	Species-typical and individual behavior in natural context	Behavioral patterns explained by mechanism, development, function, and phylogeny	Multilevel explanation; segmentation and observer dependence remain difficult.
Ecological psychology	Organism-environment coordination	Perception-action relations and affordances	Avoids isolated stimulus-response units; may underdescribe symbolic and institutional mediation.
Cognitive science	Performance generated by information-processing mechanisms	Tasks, protocols, errors, timing, internal models	Mechanistic and computational detail; latent structures can outrun evidence.
Situated/distributed cognition	Activity organized across people, artifacts, and settings	Coordinated trajectories and representational transformations	Captures context and distribution; system boundaries become analytically contested.
Cybernetics/control	Input-output relations, feedback, regulation, goal-directed dynamics	Trajectories, error signals, stability, control	General across natural and artificial systems; agency and meaning are not intrinsic.
Behavioral economics	Choices under constraints and incentives	Selections, response distributions, revealed preferences	Quantitative regularity; observed choice does not uniquely reveal preference or belief.
Reinforcement learning	Policy-generated state-action-reward trajectories	Actions, observations, rewards, returns, policies	Formal and testable; modeled state/action spaces may omit situated behavior.
Robotics	Embodied sensing, control, and action	Sensorimotor traces, trajectories, task outcomes	Links behavior to dynamics; success criteria are designer-relative.
HCI/human factors	Human performance in designed systems	Events, timing, errors, workload, task completion	Operationally rich; logs are often interface-specific and psychologically underdetermined.
Philosophy of action	Actions, agency, intention, reasons, omissions	Intentional and attributable conduct	Clarifies action/behavior distinction; less suited to non-agentive or artificial cases.

4. Canonical Distinctions

4.1 State, change, and behavior

A state is a description of a system at a time or over an interval. Behavior concerns a trajectory, transition pattern, controlled persistence, or temporally organized relation among states. Remaining still can count as behavior when maintaining position is attributable to regulation or policy; mere static existence does not. This means change is not strictly necessary, but temporal organization and counterfactual sensitivity are.

Internal state change is not automatically behavior. A memory allocation, hidden neural activation, or background process becomes behavioral evidence only relative to a declared observational level and attribution rule. Otherwise every physical change inside a system would count, eliminating useful boundaries.

4.2 Behavior and action

Behavior is the broader evidential category. Action adds an attribution of agency, organization, control, or purposive relevance. A reflex, drift, involuntary movement, policy output, and failed attempt can be behavioral without satisfying a demanding theory of action. Conversely, an omission may qualify as action when an agent could and was expected to intervene, even though no overt movement occurred. Action is therefore not reducible to a visible event.

The framework treats action attribution as typed and defeasible. Researchers must specify whether an event is merely attributed to a system, controlled by it, selected by a policy, intended, reason-guided, or normatively accountable.

4.3 Behavior and interaction

Behavior can be attributed to one system relative to an environment. Interaction requires reciprocal constraint between distinguishable systems. An agent's behavior may occur within an interaction, but not every behavior is interactive, and an interaction trace contains events attributable to multiple participants and mediating structures. Behavior is therefore a component and perspective on interaction, not its synonym.

4.4 Behavior and performance

Performance evaluates behavior against a task, norm, utility, or success criterion. The same behavioral trajectory can be high-performing under one objective and poor under another. Completion, reward, speed, accuracy, and efficiency are evaluative functions over behavior; they are not behavior itself.

4.5 Behavior and latent explanation

Intention, belief, preference, confidence, strategy, habit, expertise, attention, workload, and policy are not directly observed in ordinary traces. They are latent or theoretical variables inferred from behavior, reports, interventions, and auxiliary measurements. A scientific representation must preserve this evidential asymmetry. An action log can constrain interpretations but rarely identifies one uniquely.

5. Canonical Definition

Behavior is a bounded, temporally extended trajectory of observable or instrumentally recoverable changes, outputs, or regulated persistence attributable to a system in relation to its environment under a declared observation model.

Each clause is necessary for scientific use:

Bounded: the behaving system and relevant environment must be specified.

Temporally extended: behavior is identified through ordering, duration, transition, recurrence, or maintained regulation.

Observable or recoverable: evidence must be accessible through declared instruments or records.

Attributable: the trajectory must be assigned to a system by an explicit causal, control, ownership, or analytic rule.

Relational: behavior is interpreted relative to environmental conditions and opportunities.

Observation-model relative: event types, granularity, and uncertainty depend on how the phenomenon is measured.

This definition does not require consciousness, intention, semantic understanding, optimality, success, movement, or direct observation by a human. It also does not claim that behavior exists only when observed. Rather, scientific claims about behavior require an observation model connecting the target process to evidence.

6. Behavioral Event and Segmentation

A behavioral event is not a self-identifying atom. It is a temporally bounded segment of a trace classified under an event type and attributed to a system. Segmentation may be physical, functional, task-relative, protocol-defined, or statistically inferred. Different segmentations can be valid for different explanatory questions.

The smallest useful unit is therefore not universally a click, movement, command, or state transition. It is the minimal discriminable segment required by a declared explanatory and measurement contract. For motor control this may be a subsecond trajectory; for HCI it may be an operation; for organizational activity it may be a coordinated episode; for reinforcement learning it may be a time step; for discourse it may be a turn or act. Units must not be compared across studies without declaring their segmentation rules.

Level	Examples	Evidential status
Record	Sensor sample, log line, video frame, timestamp	Directly produced by an instrument; not yet a behavioral claim.
Detected event	Threshold crossing, message receipt, button activation	Derived from records by an explicit detector.
Attributed behavioral event	Agent selected command; robot avoided obstacle	Detected event assigned to a system and behavioral type.
Episode	Search sequence; repair attempt; conversation turn cluster	Temporally grouped events under boundary and completion rules.
Pattern	Habitual route; oscillation; repeated error	Regularity across events or episodes.
Action	Intentional, policy-selected, controlled, or normatively attributable event	Requires additional agency criteria.
Strategy	Higher-order organization across actions and contingencies	Latent or model-based; not directly read from the trace.

Level	Examples	Evidential status
Outcome/performance	Success, reward, error, cost, completion	Evaluation over traces, not behavior itself.

7. Behavior Ontology

Entity	Definition	Examples
System	Entity or analytically bounded organization to which behavior is attributed.	Human, animal, robot, software agent, group.
Environment	Conditions and entities outside the selected system boundary relevant to the behavior.	Physical setting, interface, other agents, institutional rules.
State	Declared description of system or environment at a time or interval.	Position, screen state, belief state, physiological state.
Observation model	Mapping from target processes to records and event categories.	Logging schema, coding protocol, sensors.
Record	Instrument-produced datum with provenance and timestamp.	Event log, video frame, telemetry.
Event	Segment classified under explicit detection rules.	Command invocation, gaze transition.
Attribution	Rule connecting an event or persistence to a system.	Control ownership, causal intervention, authorship.
Behavioral episode	Bounded sequence or concurrent structure of attributed events.	Search, repair, negotiation.
Context	Conditions needed to interpret or compare behavior.	Task, opportunities, policy, history, projection.
Trace	Time-indexed, partially ordered collection of records and derived events.	Session trace, multi-agent protocol trace.
Outcome	State or evaluation following a trace or episode.	Completion, failure, reward, harm.
Latent construct	Unobserved variable proposed to explain distributions of evidence.	Belief, intention, strategy, workload.
Uncertainty	Limits on detection, timing, attribution, state, or interpretation.	Confidence intervals, missing data, competing codings.

8. Behavioral Grammar

A behavioral grammar specifies admissible composition without claiming that behavior is linguistic. The minimum grammar distinguishes records, events, episodes, and explanatory annotations.

Record $r ::= \langle \text{source, time, channel, value, provenance, uncertainty} \rangle$

Event $e ::= \text{detect}(\text{type, interval, records, rule})$

Attributed event $b ::= \text{attribute}(\text{system, event, relation, confidence})$

Episode $q ::= \text{compose}(\text{boundary, order, } \{b\}, \text{context, completion})$

Trace $\tau ::= \text{partial-order}(\{\text{records, events, episodes}\}, \text{clocks, missingness})$

Interpretation $i ::= \text{infer}(\text{construct, evidence-set, model, alternatives, uncertainty})$

Outcome $o ::= \text{evaluate}(\text{trace-or-episode, criterion, tolerance})$

Composition operators include sequential composition (;), concurrency (| |), choice (+), iteration (*), interruption (!), omission ($\emptyset$ under an expectation model), and nesting ([]). These operators describe trace organization. They do not imply purpose or meaning by themselves.

9. Trace Model

A scientific trace should be modeled as more than an ordered list. Distributed and asynchronous systems require partial ordering, multiple clocks, uncertain timestamps, concurrent events, hidden state, and provenance.

Let a behavioral trace be:

$$\tau = \langle R, E, A, Q, <, C, \Omega \rangle$$

R is the set of raw records. E is the set of detected events. A maps events to attributed systems and relation types. Q is a set of episodes or higher-order segmentations. < is a temporal or causal partial order. C is context, including task, opportunity profile, projection, permissions, environment, and prior history. Ω records uncertainty, missingness, detector error, attribution confidence, and competing annotations.

A trace is evidence, not a complete account of behavior. What was not instrumented may be absent. Identical traces may arise from different mechanisms; different traces may instantiate the same strategy or task outcome. Trace equivalence must therefore be typed by observables, granularity, context, and tolerance.

10. Evidence Hierarchy

Level	Claim type	Example	Primary risk
1. Physical or digital record	Instrument output with provenance	Timestamped log, video, sensor sample	Calibration error, missingness, clock drift.
2. Derived observable	Deterministic or statistical feature from records	Duration, path length, transition count	Algorithm and threshold dependence.
3. Detected event	Segment under declared rule	Click, utterance boundary, collision	Segmentation and classifier error.
4. Behavioral attribution	Event assigned to a system	User issued command	Shared control, automation, accidental activation.
5. Episode or pattern	Composition across events	Search episode, repeated checking	Boundary choice and multiple valid decompositions.
6. Functional interpretation	Role relative to task or regulation	Error correction, exploration	Same trace may serve different functions.
7. Latent construct inference	Unobserved explanatory variable	Belief, intention, strategy, preference	Underdetermination and model dependence.
8. Causal explanation	Mechanism claimed to generate behavior	Feedback control, learning, habit	Confounding, reverse causation, omitted variables.

11. Behavior Taxonomy

Axis	Types	Question answered
Attribution	System-generated; environmentally imposed; jointly produced; uncertain	Who or what generated the trajectory?
Control	Controlled; automatic; reflexive; stochastic; externally driven	What control relation is supported?

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Axis	Types	Question answered
Temporal form	Event; persistence; sequence; cycle; concurrent coordination; developmental change	How is behavior organized in time?
Environmental relation	Reactive; anticipatory; regulatory; exploratory; communicative; transformative	How does it relate to environmental conditions?
Learning status	Novel; learned; habitual; adapted; transferred; innate/developmental	What history is evidenced?
Social organization	Individual; dyadic; collective; distributed; institutional	Where is behavioral organization located?
Outcome status	Attempted; interrupted; failed; successful; compensatory; reversible	What occurred relative to a criterion?
Observability	Directly recorded; indirectly recovered; inferred; unobserved	What evidence supports the claim?

12. Cross-System Comparative Cases

Human-computer command. A user invokes a command, the system logs an event, and a state transition follows. The record supports that an input occurred and may support attribution to the account or device. It does not alone establish intention, understanding, confidence, or authorship by the human rather than automation.

Animal navigation. A trajectory through an environment is directly behavioral under an organism boundary. Explaining it as foraging, homing, or exploration requires environmental and functional evidence. Tinbergen's distinctions show that mechanism, development, adaptive function, and phylogeny answer different questions about the same behavior.

Robot obstacle avoidance. Sensor, control, and motion traces support a mechanistic account. Goal-directedness can be specified functionally without assuming consciousness. The behavior may be generated by feedback control, a learned policy, or a fixed rule; output similarity does not identify the mechanism.

Reinforcement-learning agent. A state-action-reward trajectory is behavior under the model, while the policy is a generative disposition over actions. The environment's state representation is analytical, not necessarily the system's own representation. Reward is an evaluative signal, not evidence of subjective desire.

Distributed team activity. The relevant behavior may not be attributable to any individual. Coordination emerges across people, artifacts, and representations. A group-level episode requires explicit composition and boundary rules, preserving individual and collective levels rather than replacing one with the other.

Conversational system. Generated text is observable output. Treating it as assertion, promise, refusal, deception, or understanding requires semantic, institutional, and agency criteria. Surface linguistic form is insufficient to establish the latent construct.

13. Recurring Behavioral Structures

- Feedback loops: outputs alter conditions that shape later behavior.

- Selection under constraints: behavior is chosen or generated from an opportunity structure, whether explicitly represented or not.
- Temporal hierarchy: events compose into episodes, routines, strategies, and developmental trajectories.
- Partial observability: the behaving system and observer have incomplete and different access to state.
- Adaptation: response distributions change with history, consequences, training, and environmental structure.
- Equifinality and multifinality: different mechanisms can produce similar traces, and similar initial traces can lead to different outcomes.
- Context sensitivity: task, environment, social rules, projection, and prior activity alter the meaning of identical events.
- Coordination: behavior may be jointly organized across agents and artifacts.
- Error and recovery: failed or interrupted action changes subsequent behavior and opportunity.
- Persistence and omission: maintaining a state or not acting can be behavior under control or expectation models.

14. Latent Variables

Latent variables are legitimate when they support testable measurement models, interventions, or predictions. They become scientifically dangerous when labels merely redescribe the behavior they are supposed to explain. Saying that a user clicked because of a preference, inferred only from the click, is circular unless preference independently predicts behavior across conditions.

A latent claim should declare: construct definition; indicators; measurement model; temporal ordering; alternative explanations; identification assumptions; expected interventions; uncertainty; and falsifying observations. Self-report is additional behavior and evidence, not transparent access to an internal state. Model-generated hidden states are computational variables, not automatically psychological constructs.

The recommended policy is typed inference. Terms such as intended, believed, understood, attended, preferred, confused, and strategic must be explicitly marked as reported, experimentally manipulated, model-inferred, or theoretically attributed.

15. Behavioral Evidence Contract

Field	Required declaration
Target system	Boundary, level, and ownership criteria.
Environment	Relevant external entities, conditions, institutions, and other agents.
Observation model	Sensors, logs, coding scheme, channels, sampling, and calibration.
Temporal model	Clock, ordering, latency, synchronization, interval conventions.
Event vocabulary	Types, detection rules, segmentation, and granularity.
Attribution rule	Why an event is assigned to a system and with what confidence.
Context	Task, projection, opportunity profile, permission, history, and incentives.

Field	Required declaration
Outcome criteria	Success, failure, cost, reward, safety, or completion definitions.
Latent constructs	Measurement model, indicators, alternatives, and identification assumptions.
Uncertainty	Missing data, detector error, attribution ambiguity, and confidence.
Comparison scope	Populations, systems, settings, invariants, metrics, and tolerances.
Failure conditions	Conditions under which the trace cannot support the intended claim.

16. Falsification and Rival Accounts

Claim tested	Falsifying or contrary evidence	Consequence
Behavior can be defined as observable movement alone.	Omissions, controlled persistence, messages, and software outputs are systematically excluded.	Retain broader trajectory definition.
Behavior is simply state transition.	Transitions occur without useful attribution; a single transition omits temporal organization and environment relation.	Keep behavior as attributed trace over transitions.
Behavior is action.	Reflexes, failures, involuntary outputs, and non-agentive systems remain scientifically relevant.	Keep action as typed subset or interpretation.
Policy is behavior.	Different realized traces arise from one policy and the same trace from different policies.	Separate generative disposition from realization.
Logs directly reveal behavior.	Instrument changes, automation, shared control, and missing channels alter recorded events.	Preserve record-event-attribution hierarchy.
Latent states can be read from traces.	Multiple latent models fit the same observations or fail under intervention.	Require identified measurement and causal models.
A universal atomic unit exists.	Valid units vary by explanatory level and measurement resolution.	Use contract-relative minimal units.
Existing theories are fully sufficient without synthesis.	Cross-system comparison repeatedly conflates records, actions, states, and latent constructs.	Retain shared typed trace framework only if it improves reliability and prediction.

17. Threats to Validity

The proposed definition is intentionally scientific rather than exhaustive of ordinary language. It may classify phenomena differently from legal, moral, clinical, or folk-psychological accounts. The term attributable risks importing causal assumptions into description; this is mitigated by requiring attribution types and uncertainty.

The framework may privilege traceable computational cases. Human and animal behavior can involve embodied, affective, social, historical, and institutional structures not recoverable from event logs. Distributed cognition also makes system boundaries contested rather than simply discoverable.

Formal traces can encourage false completeness. Missing channels, observer effects, coding choices, and asynchronous processes may materially alter conclusions. Behavioral categories can become reified, particularly when inferred properties are attached to demographic groups. Culture, language,

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occupation, disability, and experience should enter through measured contexts and mechanisms rather than deterministic labels.

Finally, the framework's value is unproven until independent researchers can annotate common datasets more consistently, compare behavior across implementations, and improve explanatory or predictive performance without laundering interpretation as observation.

18. Implications for Computational Interaction Research

The framework places behavior after interaction and opportunity but before measurement and prediction. Interaction specifies reciprocal coupling. Opportunity specifies typed action availability. Behavior records the realized, attributed trajectory. Measurement establishes how traces and constructs are operationalized. Explanation connects behavior to mechanisms without assuming that representation directly causes action.

A complete analysis should therefore distinguish: what the system made possible; what was exposed or believed; what records were produced; which events were detected; which events were attributed as behavior; which actions or strategies were inferred; and which outcomes were evaluated. This separation prevents common errors such as treating an unselected control as unnoticed, a pause as confusion, task completion as understanding, or equal outcomes as equal behavior.

The model also makes later similarity research tractable. Two interactions may be semantically equivalent while producing different behavioral traces; two traces may be structurally different while instantiating the same policy; and behavior may be observationally equivalent under one instrument but distinguishable under another.

19. Conclusion

Existing behavioral theories are sufficient for their established domains but do not provide a single evidence-preserving vocabulary for computational interaction. The missing contribution is not a new substance called computational behavior. It is a disciplined synthesis separating records, events, attribution, episodes, action, performance, and latent explanation.

Behavior should be represented as a bounded, temporally extended, observation-model-relative trajectory attributable to a system in relation to its environment. This account supports humans, animals, robots, software agents, and distributed systems without requiring intention or reducing behavior to arbitrary state change. It preserves the distinction between what occurred and why it occurred.

The paper's principal conclusion is therefore: existing theories are conceptually sufficient but require a shared typed trace and evidence framework for computational interaction research. The framework should be retained only if it improves annotation reliability, cross-system comparison, causal clarity, and out-of-sample explanation or prediction.

20. Reusable Artifacts

20.1 Canonical vocabulary

Term	Definition
Behavior	A bounded, temporally extended trajectory of observable or recoverable changes, outputs, or regulated persistence attributable to a system in relation to its environment under a declared observation model.
Behavioral event	A temporally bounded segment of a trace classified under an event type and attributed to a system.
Action	Behavior attributed as controlled, selected, intentional, reason-guided, or normatively attributable under a declared theory of agency.
State	A declared description of a system or environment at a time or interval.
Trace	A time-indexed or partially ordered collection of records, detected events, attributions, episodes, context, and uncertainty.
Episode	A bounded composition of behavioral events under explicit initiation, continuation, and completion rules.
Pattern	A regularity across events or episodes under a declared similarity rule.
Performance	Evaluation of behavior relative to a task, criterion, utility, norm, or tolerance.
Latent construct	An unobserved variable proposed to explain or predict evidence through a measurement or causal model.
Behavioral evidence	Records and derived claims whose provenance, transformation, attribution, and uncertainty are declared.

20.2 Minimal annotation schema

<system_id, environment_id, record_id, source, timestamp_or_interval, event_type, detector, attribution_type, attribution_confidence, episode_id, context_id, outcome_relation, provenance, uncertainty, alternative_annotations>

20.3 Behavioral comparison protocol

- Declare systems, environments, levels, and boundaries.
- Align observation channels and verify instrumentation equivalence.
- Normalize clocks, sampling, missingness, and event-detection rules.
- Compare raw records before comparing attributed event labels.
- Declare segmentation granularity and episode boundary rules.
- Separate descriptive trace metrics from task-performance metrics.
- Compare latent constructs only through equivalent measurement models.
- Control opportunity, projection, task, permission, and prior experience.
- Use typed equivalence claims with metrics and tolerances.
- Report alternative annotations and uncertainty.

21. References

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